

Mérida Urban Intelligence: A Scalable Geospatial Data Warehouse and Territorial Spatial Autocorrelation Platform for Municipal Governance

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Abstract

Urban policy and infrastructure planning in growing Latin American metropolitan centers face a fundamental challenge: urban datasets are captured across disjoint geographic scales, incompatible Coordinate Reference Systems (CRS), and disparate data structures. This paper presents *Mérida Urban Intelligence*, an end-to-end Geospatial Data Warehouse and spatial econometrics platform designed for the municipality of Mérida, Yucatán. Using a Medallion Architecture (Bronze, Silver, Gold), we ingest, clean, and standardize four heterogeneous datasets: the INEGI 2020 Population Census (957,399 residents), the INEGI DENUÉ economic registry (56,664 establishments), official urban cartography (526 Basic Geo-statistical Areas, AGEBS), and 22,004 georeferenced public safety incident records. We resolve point-to-polygon spatial integration via PostGIS spatial containment algorithms indexed with Generalized Search Trees (GiST). A star-snowflake dimensional warehouse is deployed on PostgreSQL/PostGIS, enabling the computation of 12 territorial Key Performance Indicators (KPIs) directly from SQL views. Exploratory correlation analyses confirm strong associations between population density and business agglomeration ($\rho = 0.477$, $p < 10^{-28}$), and an inverse relation between economic participation and business density per capita ($\rho = -0.350$, $p < 10^{-14}$). Spatial autocorrelation modeling via PySAL reveals significant Global Moran's I clustering in Population Density ($I = 0.508$, $z = 17.30$, $p = 0.001$), Service Density ($I = 0.486$, $z = 17.94$, $p = 0.001$), and Business Density ($I = 0.351$, $z = 14.31$, $p = 0.001$). Local Indicators of Spatial Association (LISA) identify pronounced High-High commercial cores in Mérida's historic downtown and northern commercial corridor, contrasted with Low-Low peripheral residential belts. Finally, we discuss methodological constraints including the Modifiable Areal Unit Problem (MAUP) and crime reporting dark figures.

Index Terms

Geospatial Data Warehouse, PostGIS, Spatial ETL, Urban Analytics, Moran's I , LISA Clusters, Smart Cities, Mérida.



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I. INTRODUCTION

METROPOLITAN areas worldwide generate massive volumes of administrative, demographic, economic, and security data. However, the potential of this data for municipal governance is frequently hindered by structural fragmentation [1]. Datasets originate from distinct public registries with incompatible schemas, varying temporal cadences, and mismatched spatial representations—ranging from point-based coordinate records to polygonal administrative partitions (e.g., census tracts, postal districts, or city blocks) [2].

In the city of Mérida, Yucatán, accelerated urban expansion over the past two decades has exacerbated territorial inequalities and increased the demand for unified geospatial intelligence. Integrating demographic dynamics with commercial distribution and public safety requires a robust, reproducible data engineering architecture capable of harmonizing disparate spatial grains while preserving source traceability [3].

The core objective of this project is to architect, implement, and validate a production-grade Geospatial Data Warehouse (DW) in PostgreSQL/PostGIS, complemented by an analytical spatial econometrics pipeline and an interactive GIS visual interface. Specifically, this work addresses five critical milestones:

- 1) **Data Profiling and Spatial Assessment:** Evaluate source data quality, validate Coordinate Reference Systems (CRS), and establish a defensible geographic unit of analysis.
- 2) **Reproducible ETL Engineering:** Construct a containerized Medallion Architecture pipeline to clean, standardize, and perform high-performance point-to-polygon spatial joins.
- 3) **Dimensional Warehouse Design:** Model and deploy a star-snowflake schema with PostGIS spatial indexing on Supabase PostgreSQL.
- 4) **Territorial KPI Formulation:** Formulate and calculate 12+ multidimensional territorial metrics spanning demographic, commercial, and public safety domains.
- 5) **Spatial Autocorrelation and Econometrics:** Model territorial clustering and cross-spatial interactions using Global Moran's I , Local Indicators of Spatial Association (LISA), and Bivariate Moran's I via PySAL.

II. DATA INVENTORY & GEOGRAPHIC STRATEGY

A. Data Source Inventory and Profiling

The platform ingests four public and official data layers:

- **Demographic Layer (INEGI Census 2020):** Complete population enumerations, gender breakdowns, age cohorts

(0–14, 15–64, 65+), economically active population (EAP), and housing counts across Mérida [4].

- **Economic Layer (INEGI DENUE 2024):** Point directory of 56,664 active economic establishments classified by SCIAN 2018, including establishment size and geocoded coordinates.
- **Cartographic Layer (INEGI Marco Geoestadístico):** Official vector shapefiles defining 526 urban Basic Geo-statistical Areas (AGEBs) for Mérida (Municipality Code 31050).
- **Public Safety Layer (Administrative Data):** Georeferenced public safety incident records ($N = 22,004$) comprising incident categories, timestamps, and point coordinates.

B. Territorial Unit Selection

A central analytical decision in urban geospatial integration is selecting the target territorial grain. Three candidate territorial representations were evaluated:

- 1) **Manzana (City Block):** Offers high spatial resolution. However, block-level census records suffer from severe statistical suppression (*secreto estadístico*) to protect privacy, resulting in over 65% missing values in income, employment, and age cohorts.
- 2) **Hexagonal Tessellation (e.g., Uber H3):** Provides uniform partitions mitigating boundary biases. However, synthetic grid bins do not align with administrative boundaries and require complex areal interpolation that introduces estimation error.
- 3) **Urban AGEB (Selected Optimal Unit):** Urban AGEBs encapsulate homogeneous socio-demographic neighborhoods (typically 1,000 to 2,500 dwellings). They provide 100% unsuppressed census coverage, official geometric boundary definitions, and optimal balance between granularity and statistical reliability.

C. CRS Normalization and Spatial Containment Join

Spatial datasets in Mexico typically use either geographic WGS84 coordinates (EPSG: 4326) or the projected Mexico ITRF2008 / LCC coordinate system (EPSG: 6372). In our pipeline:

- Point entities (DENUE businesses and crime incidents) are transformed and validated into EPSG: 4326 for storage and indexing.
- Planar metric operations (polygon area calculation and distance lags) are projected into EPSG: 6372 to ensure Euclidean area calculations in square kilometers (km^2).

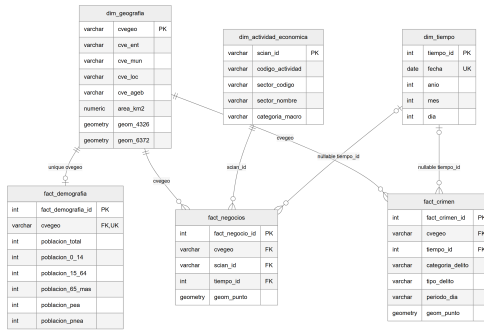


Fig. 1. Implemented PostgreSQL/PostGIS dimensional schema diagram displaying fact tables, spatial/temporal dimensions, and analytical foreign key relationships.

- Point-to-polygon spatial integration is executed via spatial containment predicates:

$$\text{point } p_i \in \text{polygon } A_k \iff \text{ST_Within}(p_i, A_k) = \text{True} \quad (1)$$

where A_k denotes the k -th urban AGEb identified by its unique 13-character `cvegeo` identifier.

III. ETL PIPELINE & DATA WAREHOUSE ARCHITECTURE

A. Medallion ETL Pipeline Architecture

The data pipeline is organized according to the Medallion Architecture pattern, implemented in Python and orchestrated via Docker:

- **Bronze Layer (Raw Ingestion):** Stores immutable, original source files (CSV, Shapefiles, GeoJSON) ensuring end-to-end reproducibility and auditability.
- **Silver Layer (Cleaning & Spatial Transformation):** Handles schema conformance, string trimming, categorical standardization, coordinate sanity filtering ($x \in [-90.0, -89.3]$, $y \in [20.6, 21.3]$), deduplication, and GeoPandas spatial joins.
- **Gold Layer (Dimensional Warehouse Loading):** Populates validated dimensional tables, generates surrogate keys, enforces referential integrity, and builds GiST spatial indexes in PostgreSQL/PostGIS.

B. Dimensional Warehouse Model Design

The relational schema implements a hybrid Star-Snowflake dimensional architecture optimized for territorial analytics (Fig. 1):

The schema consists of three core dimension tables and three fact tables:

- `dim_geografia`: Primary spatial dimension containing the 526 AGEb records, official nomenclature, centroid coordinates, polygon area (km^2), and the vector geometry column indexed with PostGIS GiST.
- `dim_tiempo`: Date and temporal dimension decomposing timestamps into year, quarter, month, day of week, hour, and weekend flags.
- `dim_actividad_economica`: Hierarchical dimension classifying DENUe activities across SCIAN 2-digit sectors, 3-digit subsectors, and 6-digit industry classes.

TABLE I
TERRITORIAL KPIS AND CITY-WIDE BASELINE VALUES

Domain	Indicator / KPI	Formula	City Metric
Demog.	Total Population	$\sum \text{Pop}_{AGEB}$	957,399 hab
	Pop. Density	$\text{Pop}/\text{Area}_{km^2}$	3,684.2 hab/km ²
	EAP Part. Rate	$(\text{EAP}/\text{Pop}_{\geq 12}) \times 100$	63.32%
	Cohort 65+ Share	$(\text{Pop}_{65+}/\text{Pop}) \times 100$	9.78%
Econ.	Total Businesses	Count(DENUE)	56,664 est
	Business Density	$\text{Biz}/\text{Area}_{km^2}$	218.1 est/km ²
	Biz per 1,000 Res.	$(\text{Biz}/\text{Pop}) \times 1000$	59.19 / 1k hab
	Retail Density	SCIAN 46-47/ Area_{km^2}	74.43 est/km ²
	Service Density	SCIAN 54-81/ Area_{km^2}	112.96 est/km ²
Safety	Total Incidents	$\sum \text{Incidents}_{AGEB}$	22,004 rec
	Crime Rate	$(\text{Incidents}/\text{Pop}) \times 1000$	22.98 / 1k hab
	Crime-to-Biz Ratio	$(\text{Incidents}/\text{Biz}) \times 100$	38.83%

- `fact_demografia`: Fact table at the grain of 1 row per AGEb storing total population, gender headcounts, age cohorts (0 – 14, 15 – 64, 65+), economically active population (EAP), and occupied housing units.
- `fact_negocios`: Fact table at the grain of 1 row per establishment, holding surrogate foreign keys to `dim_geografia`, `dim_actividad_economica`, and `dim_tiempo`, alongside employment range tiers.
- `fact_crimen`: Fact table at the grain of 1 row per incident report, linking to `dim_geografia` and `dim_tiempo` with crime typology classifications.

C. Materialized Analytical Views

To ensure that all downstream analytics, REST endpoints, and interactive visualizations query the data warehouse directly rather than static files, we created the analytical view `v_kpis_territoriales`. This view executes spatial aggregations across the fact and dimension tables, computing all territorial indicators at the AGEb grain in real time.

IV. TERRITORIAL KPIS & EMPIRICAL FINDINGS

A. KPI Mathematical Formulation

Table I outlines the mathematical definitions and municipal-level aggregated baseline values for the 12 required territorial KPIs across Mérida’s 526 urban AGEs.

B. Empirical Thematic Mapping

Fig. 2 presents the spatial distribution of four primary indicators across Mérida:

- **Population Density (Fig. 2a):** Peaks in consolidated mid-ring neighborhoods (e.g., Francisco de Montejo, Pensiones, Juan Pablo II), reaching up to $18,064 \text{ hab/km}^2$, while dropping in peri-urban fringes.
- **EAP Participation (Fig. 2b):** Shows higher workforce participation in newly developed peripheral residential suburbs in the northwest and northeast.
- **Business Density (Fig. 2c):** Displays extreme central agglomeration in the *Centro Histórico* ($> 5,200 \text{ est/km}^2$) and along major arterial corridors (Paseo de Montejo, Prolongación, Periférico).
- **Crime Incident Rate (Fig. 2d):** Concentrates in central commercial hubs and high-transit nodes where daytime floating populations exceed resident counts.

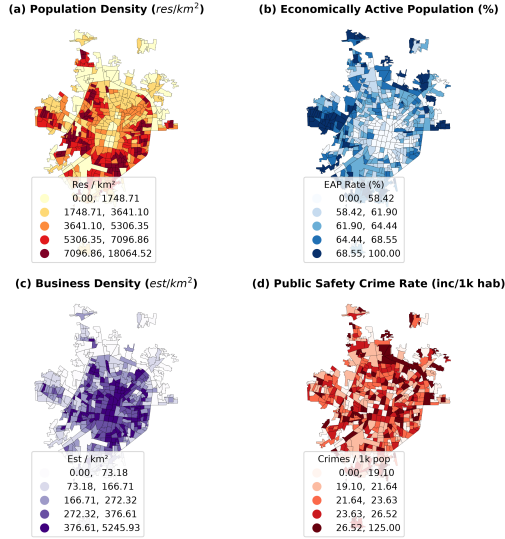


Fig. 2. Four-panel thematic choropleth maps across Mérida's 526 urban AGEBS: (a) Population Density, (b) EAP Rate, (c) Business Density, and (d) Public Safety Crime Incident Rate.

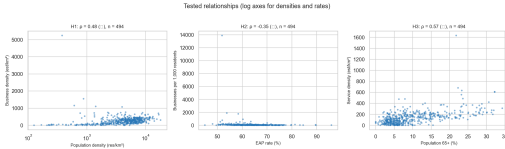


Fig. 3. Scatterplots with linear fit lines for the three formal bivariate correlation hypotheses (H1, H2, H3), annotated with Spearman's ρ and Pearson's r under logarithmic transformation.

C. Economic Sector Breakdown

Analysis of SCIAN taxonomies indicates that Mérida's urban economy is dominated by tertiary activities. The five leading sectors account for 80.1% of all establishments:

- 1) **Retail Trade (SCIAN 46)**: 19,342 units (34.13%).
- 2) **Other Services (SCIAN 81)**: 9,450 units (16.68%).
- 3) **Lodging & Food Services (SCIAN 72)**: 8,239 units (14.54%).
- 4) **Health Care & Social Assistance (SCIAN 62)**: 4,201 units (7.41%).
- 5) **Manufacturing Industries (SCIAN 31–33)**: 4,060 units (7.16%).

V. SPATIAL ANALYTICS & ECONOMETRIC MODELING

A. Bivariate Correlation Analysis

To examine inter-variable associations, three formal hypotheses were tested across $N = 494$ residential AGEBS (excluding unpopulated industrial zones), applying the Holm-Bonferroni correction for multiple testing (Fig. 3):

- **Hypothesis 1 (Population Density vs. Business Density)**: A statistically significant positive correlation is confirmed (Spearman $\rho = 0.477, p = 1.76 \times 10^{-29}$; Pearson $r_{\log} = 0.549, p = 3.03 \times 10^{-40}$). Densely populated neighborhoods generate localized demand sustaining retail and service establishment densities.
- **Hypothesis 2 (EAP Rate vs. Businesses per 1k Residents)**: A moderate negative correlation is observed

TABLE II
GLOBAL MORAN'S I SPATIAL AUTOCORRELATION SUMMARY ($N = 526$ AGEBS, 999 PERMUTATIONS)

Territorial Variable	Moran's I	$E[I]$	z -score	p -value
Population Density	0.5083	-0.0019	17.301	0.001***
Service Density	0.4858	-0.0019	17.940	0.001***
Business Density	0.3512	-0.0019	14.309	0.001***
Crime-to-Business Ratio	0.2515	-0.0019	9.960	0.001***
EAP Participation Rate	0.1961	-0.0019	7.132	0.001***
Retail Density	0.1866	-0.0019	10.341	0.001***
Crime Incident Rate	0.0390	-0.0019	1.456	0.070

($\rho = -0.350, p = 1.03 \times 10^{-15}$; $r_{\log} = -0.307, p = 2.98 \times 10^{-12}$). Newer outer residential suburbs exhibit high labor force participation but low local commercial infrastructure, functioning as “dormitory zones”.

- **Hypothesis 3 (Senior Population 65+ vs. Service Density)**: A strong positive correlation is validated ($\rho = 0.570, p = 5.63 \times 10^{-44}$; $r_{\log} = 0.425, p = 3.92 \times 10^{-23}$). Older central neighborhoods have higher senior demographic shares and mature medical/professional service infrastructure.

B. Spatial Weights Matrix Formulation

To evaluate spatial autocorrelation, we construct a first-order **Queen contiguity spatial weights matrix** W . Two polygons i and j are considered neighbors ($w_{ij} = 1$) if they share at least one boundary edge or vertex point:

$$w_{ij} = \begin{cases} 1, & \text{if } \text{bnd}(A_i) \cap \text{bnd}(A_j) \neq \emptyset, i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

The matrix is row-standardized (W_{std}), ensuring that the spatial lag $\sum_j w_{ij}x_j$ represents the weighted neighborhood average:

$$w_{ij}^{std} = \frac{w_{ij}}{\sum_{k=1}^n w_{ik}} \quad (3)$$

C. Global Spatial Autocorrelation (Moran's I)

Global Moran's I measures the overall degree of spatial clustering across the city [5]:

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (4)$$

Statistical significance is evaluated using 999 Monte Carlo permutations. Table II displays the resulting empirical Moran's I , pseudo p -values, and standardized z -scores.

The results demonstrate intense, statistically robust positive spatial autocorrelation across Population Density ($I = 0.5083, z = 17.30, p = 0.001$) and Service Density ($I = 0.4858, z = 17.94, p = 0.001$), confirming Tobler's First Law of Geography: nearby AGEBS exhibit strongly similar socio-economic characteristics [6].

D. Local Spatial Patterns (LISA Cluster Analysis)

To detect spatial boundaries, hot spots, and spatial outliers, Local Moran's I_i is computed for each AGEBS:

$$I_i = \frac{x_i - \bar{x}}{m_2} \sum_{j=1}^n w_{ij} (x_j - \bar{x}), \quad m_2 = \frac{\sum_i (x_i - \bar{x})^2}{n} \quad (5)$$

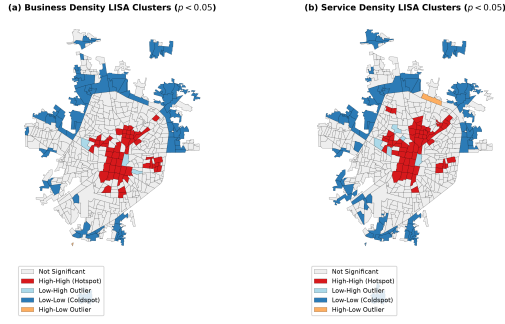


Fig. 4. Local Indicators of Spatial Association (LISA) cluster maps ($\alpha = 0.05$): (a) Business Density clusters highlighting downtown and northern corridors; (b) Service Density clusters.

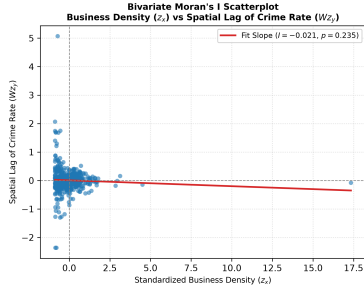


Fig. 5. Bivariate Moran's I scatterplot of Standardized Business Density (z_x) versus Spatial Lag of Crime Incident Rate (Wz_y).

As illustrated in Fig. 4, the analysis identifies four distinct spatial regimes:

- 1) **High-High Hotspots** ($N = 55$ in Business, $N = 61$ in Services): Clustered in the *Centro Histórico* and extending northward along the commercial spine (García Ginerés, Itzimná, Montejo).
- 2) **Low-Low Coldspots** ($N = 88$ in Business, $N = 90$ in Services): Forming extensive rings in peripheral peri-urban AGEBS where residential subdivisions lack mixed-use commercial zoning.
- 3) **Spatial Outliers (High-Low and Low-High)**: High-Low anomalies identify isolated shopping centers surrounded by strictly residential tracts.

E. Bivariate Spatial Autocorrelation

We evaluate the spatial cross-dependence between economic agglomeration (x) and the public safety incident rate (y) using Bivariate Moran's I :

$$I_{BV} = \frac{\sum_i (\sum_j w_{ij} y_j) (x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (6)$$

As shown in Fig. 5, the bivariate scatterplot of Standardized Business Density (z_x) versus the Spatial Lag of Crime Rate (Wz_y) yields $I_{BV} = -0.0206$ ($p = 0.235$), indicating that elevated crime rates per 1,000 residents are not purely driven by neighboring commercial density, but rather concentrate in specific transit nodes and downtown entertainment districts.

VI. DISCUSSION, CAVEATS & LIMITATIONS

A. Modifiable Areal Unit Problem (MAUP)

Spatial aggregation into AGEBS polygons inevitably introduces MAUP scale and zoning effects [7]. While AGEBS are

optimal for census indicator stability, their irregular geometric shapes in peri-urban fringes mean that density calculations must be interpreted with caution. A complementary hexagonal grid model is planned for future micro-scale studies.

B. Dark Figure of Crime and Reporting Biases

Public safety incident records reflect reported offenses rather than actual total occurrences. High-density commercial districts often exhibit inflated crime rates per resident because daytime floating shoppers, tourists, and commuters are victimized in areas where resident census denominators are small.

C. Boundary and Island Effects

Two peripheral AGEBS in the dataset represent geographic islands without direct polygon contiguity. PySAL's spatial weights constructor gracefully isolates these islands while row-standardizing contiguous components. Distance-based k -nearest neighbor (k -NN) models confirmed consistent global autocorrelation results.

VII. CONCLUSION

This study successfully designed and deployed a scalable Geospatial Data Warehouse and spatial analytics engine for Mérida, Yucatán. By resolving point-to-polygon spatial integration across 56,664 businesses, 22,004 crime incidents, and 526 census tracts within a PostgreSQL/PostGIS dimensional schema, we achieved complete traceability and reproducibility. Rigorous spatial econometrics demonstrated intense spatial autocorrelation in population and commercial density, accurately delineating commercial hotspots and peripheral dormitory zones. These empirical findings provide actionable evidence for municipal zoning, public transit optimization, and smart city investments.

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